

Exact Long-Twist Recurrences and Streamed Homology

for Unknot Recognition

Finite thresholds, compressed braid inputs, measured implementation tradeoffs, and the remaining quasi-polynomial problem

Prepared for Vladimir Reshetnikov's ProveIt project

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Abstract

We continue the integrated twist-compression and component-algebra work in ProveIt at revision `ea2abcb115aaa58f0b193ce1e045c2def983e1e6`. For a signed braid with one variable run $\sigma_i^{\varepsilon m}$, freeze the other runs and let their total crossing count be W . We prove that one reduced characteristic-two Khovanov computation at $m_0 = W+2$ determines the entire homological-degree profile for every $m \geq m_0$. Two fixed finite boundary pieces are separated by a constant-rank interval; the exact rank is

$$R(m) = R(m_0) + (m - m_0)\eta, \quad \eta = \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(J_E; \mathbb{F}_2) > 0,$$

where J_E is the closure with the selected run replaced by its turnback smoothing. The proof uses the repeated differential $D_E + X_p + X_q$, not just repeated chain dimensions. Jaeger's weighted-dot theorem identifies the slope. A compact output makes the exponent dependence polynomial in its bit length for fixed context.

We implement the recurrence, a degree-streamed exact backend with bounded state and map caches, and direct structural certificates for run-encoded braids. Paired experiments show substantial standalone homology gains from the recurrence and a measurable time-space tradeoff from streaming. Independent crossing-cube comparisons check both signed gradings and the full homological profiles. We also prove that a proposed definite-signature shortcut is dominated by the existing Rasmussen interval, and that this interval already decides every knot in the recurrence's dominant-run range. Thus the new exact-homology result does not enlarge that decision class. The work gives precise restricted-input bounds and an implementable research programme; it does not establish unrestricted quasi-polynomial recognition.

Deliverable: a reproducible article, additive implementation, regression tests, independent audits, raw measurements, and a patch against the pinned baseline. Mathematical proofs and executable checks are distinguished throughout.

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1 Conclusions, baseline, and research scope

The strongest new mathematical step in this continuation is an exact finite-threshold recurrence for one long twist region. The existing support-counting formula says that the chain dimension grows affinely in a selected exponent. We pass from that statement about the number of generators to a statement about the differential and hence the actual homology. This distinction matters: a repeated vector space can carry different differentials, and chain-size growth alone does not determine homology growth.

The implementation work has two complementary purposes. The recurrence avoids constructing an arbitrarily long repeated part. The streamed backend reduces retained storage when the finite complex still needs to be evaluated. Direct braid-source structural bounds avoid expanding a binary exponent merely to rediscover a certificate already available from elementary counts. Each change has an explicit interface and an independently stated scope.

1.1 What is already in ProveIt

The baseline is the repository’s main revision `ea2abcb115aaa58f0b193ce1e045c2def983e1e6`, retrieved on 8 October 2026. The immediately relevant integrated work comprises report 10 on structural and algebraic improvements, report 11 on twist compression, report 12 on component algebra, and their synthesis updates [1]. The complete `fast/` Python source, tests, examples, and build metadata at this revision accompany the package as a reference snapshot.

The current recognizer already has much more than a naive crossing cube. Its pipeline contains checked diagram construction, braid certificates including the at-most-three-strand endpoint, signed Seifert-graph criteria, diagram simplification, visible connected-sum and interlacement factorization, modular Alexander and Jones obstructions, and exact Khovanov scanning. Optional paths provide a twist macro backend and component-algebra contractions. Existing scan machinery includes local gluing, unit cancellation, bit-packed linear algebra, and choices of scan order. Those features are baseline capabilities, not contributions of this continuation.

In particular, report 11 explicitly proposed degree-streamed rank computation and required the state generator and map caches to be included in its memory analysis. We implement that proposal rather than presenting the rank identity underlying it as a new mathematical discovery. Report 11 also established, for supplied signed runs with lengths m_j , the support polynomial

$$\mathcal{N} = \sum_{S \subseteq [t]} 2^{c(S)-1} \prod_{j \in S} m_j \quad (1)$$

and a restricted-family complexity bound

$$\text{poly}(n) 2^{O(t \log(2+n/t))}, \quad n = \sum_j m_j. \quad (2)$$

Here $c(S)$ is the number of circles after choosing the turnback exactly in the boxes in S . The run count t is the count in the supplied presentation. It is not a proven small invariant of every knot.

Result or capability	Status in this continuation
Two-strand twist normal form; Khovanov unknot detector	Established literature, used with explicit attribution.
Support polynomial and $t = O(\log n)$ bound	Already integrated in the pinned ProveIt baseline.
One-reference signed homology recurrence and slope extraction	Proved here as a continuation of that model; implemented.
Slope as reduced homology of the turnback closure	Proved using established dot-sliding and local contraction machinery.
Degree-streamed ranks, bounded caches, early lower bound	Implemented with exact accounting and regression checks.
Structural certificates on binary run input	An encoding improvement to existing sufficient conditions.
New signature packing decision class	None: the proposed bound is proved dominated.
Unrestricted quasi-polynomial recognition	Not established by this work.

1.2 Current literature and the asymptotic target

The mathematical local input is Bar–Natan’s tangle category and cancellation calculus [2, 3], with integer-twist complexes described explicitly by Thompson [4]. The rank-one decision criterion is due to Kronheimer and Mrowka [7]. Twist stabilization and compact tangle descriptions have substantial precedent, including the immersed-curve framework of Kotelskiy, Watson, and Zibrowius [6]. The finite threshold and one-reference extraction below should be assessed as a precise, executable continuation of the repository’s model. We make no general priority claim for stability under twisting.

Kelomäki and Schütz’s 2026 preprint supplies an especially useful boundary for complexity arguments: their Theorem 1.2 gives exponential scanning behavior on certain three-strand braid families, whereas their Theorem 1.3 gives a specialized polynomial-time computation for all three-strand braids [8]. A bounded number of boundary matching types therefore cannot, by itself, bound the number of algebraic summands. The specialized theorem also prevents a misleading benchmark claim: three-strand examples are useful algebraic checks, but do not demonstrate new difficult instances for the integrated recognizer.

The hierarchy route must be described separately. Lackenby’s July 2026 preprint develops incompressible surfaces, boundary patterns, and unknot recognition. Section 9 bounds iterations while explicitly leaving costs of steps insufficiently specified for a complete runtime estimate; Proposition 9.1 gives $L(g+1)^L$ iterations in its hierarchy and pattern parameters [13]. Other results provide compact surface certificates and algorithms polynomial in handle data and logarithmic normal weight. Those facts support a serious alternative research programme, but they do not supply the missing cost theorem for the present Khovanov implementation. Certificate existence, certificate verification, certificate search, and repeated compressed operations remain different tasks.

We also inspected the user-suggested `openai/math` catalogue and repository guidance [14]. Its manuscripts have varying verification status. No theorem from that collection is needed for the proofs here; a superficially similar quasi-polynomial result in a different subject would not remove the particular representation and rank barriers identified below. The useful methodological lesson is to keep dependencies, proof status, and machine-checkable artifacts explicit.

1.3 Four sizes that must be kept distinct

Let

$$\beta = \sigma_{i_1}^{e_1} \cdots \sigma_{i_t}^{e_t}, \quad 1 \leq i_j < b, \quad e_j \in \mathbb{Z} \setminus \{0\}. \quad (3)$$

Put $m_j = |e_j|$, $n = \sum_j m_j$, and define the encoded input size, up to a fixed choice of delimiters, by

$$\lambda = 1 + \lceil \log_2(b+1) \rceil + \sum_{j=1}^t (1 + \lceil \log_2(b+1) \rceil + \lceil \log_2(m_j+1) \rceil). \quad (4)$$

An expanded braid word has length n . A run-encoded braid can have n exponential in λ . A planar diagram normally presents its crossings explicitly and therefore has a different input model.

There are also the number $S = \prod_j (m_j + 1)$ of macro states, the total reduced chain dimension $\mathcal{N}$, and the bit length of the requested output. A scalar formula for $\mathcal{N}$ does not compute a differential rank. A compact formula for a homological profile is a different output from an explicit dictionary containing each degree. We charge for the chosen representation throughout.

The closure of a braid is a knot exactly when its strand permutation is a single cycle. A run contributes the adjacent transposition only when its exponent is odd. Thus original component count is computable without expanding any run. For a one-component b -braid every generator index $1, \dots, b-1$ occurs; omitting index i separates the first i strands from the others. This simple fact gives $b-1 \leq t$ and will also control the reference size in the dominant-run theorem.

1.4 The characteristic-two exact endpoint

For a knot K , the endpoint used here is

$$K \text{ is the unknot} \iff \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2) = 1. \quad (5)$$

The implication from rank one follows from the rational rank-one detector of Kronheimer–Mrowka and the universal coefficient theorem: characteristic two dimension is at least the rational dimension, and reduced homology is nonzero. The unknot has one reduced generator over either field. All calculations retain a fixed reduction point. For links we compute reduced homology at that point, but do not apply a knot-only decision criterion to them.

Resource exhaustion has a precise meaning. A mathematical exact algorithm without fixed caps terminates on finite inputs. Its resource-bounded implementation can return UNKNOWN. Such a result is neither evidence of knottedness nor evidence of triviality. The same distinction applies when a computation has established only a lower bound on the homology dimension.

2 The local complex and its global counts

We use $A = \mathbb{F}_2[X]/(X^2)$, with

$$\Delta(1) = 1 \otimes X + X \otimes 1, \quad \Delta(X) = X \otimes X.$$

A resolution with c circles has reduced space of dimension 2^{c-1} : the circle containing the fixed mark carries X , and each other circle carries 1 or X . Let I denote the identity two-strand matching and E the matching joining the two top ends and joining the two bottom ends. Names such as cap and cup depend on drawing orientation; the chosen arcs are always identified as the top and bottom arcs of E .

Proposition 2.1 (Established local normal form). *After forgetting quantum grading, a positive run of length m is represented by*

$$I \xrightarrow{s} E \xrightarrow{L} E \xrightarrow{L} \cdots \xrightarrow{L} E, \quad L = X_{\text{top}} + X_{\text{bottom}},$$

in degrees $0, 1, \dots, m$. A negative run is represented by its dual complex in degrees $-m, \dots, -1, 0$, with I in degree zero, the final saddle into I , and the same internal dot sum. The replacement is valid inside any fixed exterior tangle.

Proof. This is the characteristic-two, quantum-forgotten specialization of the integer-twist normal form [4, Proposition 4.1]. Tensoring with one additional crossing gives an E -circle summand. Delooping that circle and cancelling an invertible saddle block gives the next zig-zag term. The local contraction formula is the categorical Schur complement of Bar–Natan [3]. Over the integers the internal maps have alternating signs; the dot sum and difference agree over $\mathbb{F}_2$. Mirroring reverses degrees and dualizes saddles. These operations are relative to the boundary, which justifies the stated exterior compatibility. $\square$

In the tensor product of run complexes, a state has coordinates $0 \leq k_j \leq m_j$. Its degree is $\sum_j \text{sign}(e_j)k_j$. Its geometry depends only on the support $\{j : k_j \neq 0\}$, while its differentials also distinguish the first saddle from internal dot maps. For a fixed support S , the number of coordinate choices is $\prod_{j \in S} m_j$. This proves (1). The independence of geometry from the nonzero coordinate is also the key to the stronger homology result below.

For later accounting define

$$d_h = \dim C^h, \quad r_h = \text{rank}(\partial_h : C^h \longrightarrow C^{h+1}), \quad \mathcal{N} = \sum_h d_h. \quad (6)$$

The exact formula

$$\beta_h = \dim H^h = d_h - r_{h-1} - r_h \quad (7)$$

uses both incident boundary ranks. Computing only chain dimensions, only one boundary rank, or a rank bound in the wrong direction cannot justify an exact Betti number.

Two convenient coarse quantities are

$$Q = \sum_h d_h d_{h+1}, \quad T_{\text{elim}} = O\left(\mathcal{N} + \sum_h d_h d_{h+1} \min(d_h, d_{h+1})\right)$$

for elimination once the actual columns are supplied. Generating states and local maps is a separate cost; it is included explicitly in the algorithm bounds below. The $\mathcal{N}$ term also charges for visiting zero columns and end degrees. Q is a dense bit envelope, not a count of actual nonzero entries. The displayed elimination term is a conservative bit-operation bound for ordinary characteristic-two Gaussian elimination. It is enough for asymptotic upper bounds; it is not a claim that Python attains an optimal matrix multiplication exponent.

3 An exact finite-threshold recurrence

Select one run position in (3), and keep its generator, its sign ε , and every other run fixed. Only its absolute exponent $m \geq 1$ varies. Define

$$a = - \sum_{\substack{j \neq * \\ e_j < 0}} |e_j|, \quad c = \sum_{\substack{j \neq * \\ e_j > 0}} |e_j|, \quad W = c - a, \quad m_0 = W + 2. \quad (8)$$

The letter c here is a degree endpoint, not a circle count or the number b of braid strands. Empty sums are zero.

3.1 The fixed context and its repeated differential

For each context degree q , let I^q be the direct sum of reduced resolution spaces when the distinguished box contains I , and let E^q be the corresponding sum with E in that box. Both vanish outside $[a, c]$. All other run differentials define D_I and D_E . The chosen saddle maps between the two sectors in the direction determined by ε . Write

$$V = \bigoplus_{q=a}^c E^q, \quad M = \dim V, \quad L = X_p + X_q, \quad F = D_E + L, \quad r = \text{rank } F, \quad \eta = M - 2r. \quad (9)$$

In $X_p + X_q$, the subscripts designate the two exterior arc points, independently of the context degree index. The operator D_E raises context degree, while L preserves it. Consequently F is initially an endomorphism of the *ungraded* space V .

Lemma 3.1. *The operator F satisfies $F^2 = 0$, and $\eta = \dim H(V, F) \geq 0$.*

Proof. The context is a chain complex, so $D_E^2 = 0$. Each dot squares to zero, and distinct dots commute; hence $L^2 = 0$ in characteristic two. Multiplication and comultiplication commute with a dot on an unchanged arc:

$$m(X \otimes 1) = Xm = m(1 \otimes X), \quad (X \otimes 1)\Delta = \Delta X = (1 \otimes X)\Delta.$$

These are direct identities in A , including on the marked- X subcomplex. Thus $D_E L = L D_E$, and the mixed terms in F^2 cancel. Finally, $\text{im } F \subseteq \ker F$; rank-nullity gives $\dim \ker F - \dim \text{im } F = M - 2r$. $\square$

The identity $F^2 = 0$ is stronger than a numerical observation about consecutive ranks. Its use is specific to the characteristic-two normal form just specified. An integral or odd-characteristic version needs its own signs, gradings, and differential analysis.

3.2 Where complete context groups occur

For a positive distinguished run, the summand E^q lies in total degree $h = q + k$, with $1 \leq k \leq m$. Therefore

$$C_m^h = V \quad \text{throughout} \quad c + 1 \leq h \leq a + m. \quad (10)$$

Indeed, these inequalities are exactly the conditions that every $q \in [a, c]$ has $1 \leq h - q \leq m$. The I sector is absent in this interval because $h > c$. Between two adjacent groups satisfying (10), the other-run maps give D_E and the distinguished internal maps give L . Under the canonical identification by context state and circle label, the differential is exactly F .

For a negative distinguished run, $h = q - k$ instead. Complete groups and their internal differentials occur in

$$c - m \leq h \leq a - 1, \quad \partial_h = F \quad (c - m \leq h \leq a - 2). \quad (11)$$

Here the I sector is absent because $h < a$. These descriptions keep the mixed signs in the context; no assumption that the braid is positive, alternating, or homogeneous has entered.

At $m = m_0$, there are exactly two consecutive complete groups. In the positive case they are in degrees $c + 1, c + 2$; in the negative case they are in degrees $a - 2, a - 1$. The single map between them is F . The two-group threshold is deliberate. One complete group alone does not expose a complete central differential, and its Betti number can still depend on both boundary regions.

3.3 The two signed profile formulas

Let

$$P_m(z) = \sum_h \dim_{\mathbb{F}_2} H^h(C_m) z^h$$

be the reduced homological Laurent polynomial, with quantum grading forgotten and the macro grading fixed as in proposition 2.1.

Theorem 3.2 (One-reference recurrence). *Set $m_0 = W + 2$, $\delta = m - m_0$, and compute P_{m_0} . For a positive distinguished run, split it into*

$$P_{\text{low}} = \sum_{h \leq c+1} \beta_h(m_0) z^h, \quad P_{\text{high}} = \sum_{h \geq c+2} \beta_h(m_0) z^h.$$

Then for every $m \geq m_0$,

$$P_m(z) = P_{\text{low}}(z) + z^\delta P_{\text{high}}(z) + \eta \sum_{h=c+2}^{a+m-1} z^h. \quad (12)$$

For a negative distinguished run, put

$$Q_{\text{low}} = \sum_{h \leq a-2} \beta_h(m_0) z^h, \quad Q_{\text{high}} = \sum_{h \geq a-1} \beta_h(m_0) z^h.$$

Then

$$P_m(z) = z^{-\delta} Q_{\text{low}}(z) + Q_{\text{high}}(z) + \eta \sum_{h=c-m+1}^{a-2} z^h. \quad (13)$$

An interval sum with lower endpoint above its upper endpoint is empty. In either case the total reduced rank obeys

$$R(m) = R(m_0) + (m - m_0)\eta. \quad (14)$$

The statement permits links and any fixed exterior reduction point.

Proof. First suppose the distinguished run is positive. Near the lower end, the condition $k \leq m$ is inactive once $m \geq m_0$. Every chain group in degrees at most $c + 1$, its incoming map, and its outgoing map are therefore the same as at m_0 . At the top, writing $k' = k - \delta$ identifies all groups in degrees at least $c + 2 + \delta$, together with both incident maps, with the reference upper region translated by δ . This can also be read directly from $1 \leq h - q \leq m$: near that end the lower bound is inactive. The saddle into the first E sector remains in the fixed lower region.

The degrees strictly between these two regions are $c + 2, \dots, a + m - 1$. Each has chain space V , incoming map F , and outgoing map F . Its Betti number is $M - r - r = \eta$. There are $m - m_0$ such degrees. Applying (7) in the fixed and translated boundary regions proves (12).

For a negative run, use $1 \leq q - h \leq m$. The lower region now translates downward by δ , while the upper region, including the saddle into I , remains fixed. Its separating complete groups are (11). The inserted degrees are precisely $c - m + 1, \dots, a - 2$, again with both maps F . This proves (13). Evaluating at $z = 1$ gives (14). $\square$

This proof identifies the boundary maps as well as the central chain groups. Without that step, shifting observed Betti numbers would be an unjustified extrapolation. It also explains why the reference can be a link of different component count: the argument is a statement about marked complexes, not an isotopy between the original and reference closures.

Corollary 3.3 (Extraction from existing output). *The ordinary macro output at m_0 , which reports chain dimensions and boundary ranks, determines η without a second homology computation. Use*

$$h_* = \begin{cases} c + 1 & \varepsilon = +1, \\ a - 2 & \varepsilon = -1, \end{cases} \quad M = d_{h_*} = d_{h_*+1}, \quad r = r_{h_*}, \quad \eta = M - 2r.$$

The threshold Betti numbers at h_* and $h_* + 1$ need not equal η . They are boundary-region values. This distinction is covered by tests comparing the extracted slope with the newly interior degree at $m_0 + 1$, in addition to tests of total ranks.

3.4 Chain savings and a concrete family

Let $A_0 = \sum_q \dim I^q$, $B_0 = \sum_q \dim E^q = M$. The original and reference chain dimensions are exactly

$$\mathcal{N}(m) = A_0 + mB_0, \quad \mathcal{N}_0 = A_0 + (W + 2)B_0. \quad (15)$$

The coefficient B_0 counts repeated chain generators. The coefficient $\eta = B_0 - 2 \operatorname{rank} F$ counts repeated homology. They are usually different.

For example, take

$$\beta_m = \sigma_1^m \sigma_2^{-1} \sigma_1 \sigma_2^{-1} \quad \text{on three strands,} \quad m \geq 5.$$

Here $(a, c, W, m_0) = (-2, 1, 3, 5)$. The reference computation gives $\mathcal{N}_0 = 93$, $B_0 = 15$, $r = 6$, and $\eta = 3$, while $\mathcal{N}(m) = 15m + 18$ and $R(m) = 3m + 2$. The result applies to every integer $m \geq 5$, including link parities; odd m gives a knot. The figure included with the experiments plots the full degree profile and its translated upper boundary. This is a useful exact test family, not a hard recognition family: the baseline already has structural and three-strand certificates.

4 The slope is the homology of the smoothing closure

The finite-threshold recurrence identifies its slope as the homology dimension of a repeated differential. We now identify that number topologically. This also proves that the repeated part cannot disappear. All vector spaces in this section are over $\mathbb{F}_2$, and the reduction point is fixed on a closing arc outside the run boxes. Let

$$V = \bigoplus_q E^q, \quad F = D_E + L, \quad L = X_p + X_q, \quad \eta = \dim V - 2 \operatorname{rank} F,$$

where p and q lie on the cap and cup of the distinguished E smoothing. Here D_E is the context differential. The notation X_p denotes the usual dot action, namely multiplication by X on the resolution circle containing p . Let J_E be the actual link diagram obtained by replacing the entire distinguished run by that smoothing. The identification below concerns total homology dimension; it does not identify the original homological grading on the summands of V with a grading on $H(V, F)$.

4.1 The two dots lie on one actual component

Lemma 4.1. *The cap and cup of the distinguished smoothing belong to the same component of J_E , for every choice of the other signed braid runs.*

Proof. Remove the interior of the smoothing box and retain the four cut ends. The outside diagram consists of closed components and two open paths. The braid orientation implies that each open path joins a top cut end to a bottom cut end: two ends of the same kind cannot be the ends of a coherently oriented outside path. Consequently the outside matching of these four ends is one of the two top-to-bottom matchings. Joining the two top ends by the cap and the two bottom ends by the cup produces a single closed component in either case. This component contains both smoothing arcs. The argument concerns the actual strands, with crossings retained. In a complete resolution the two arcs can lie on different circles, so it would be incorrect to set $L = 0$ state by state. $\square$

4.2 The characteristic-two weight slide

Jaeger's weighted-dot construction proves that, in characteristic two, relocating markings while preserving their total weight on every actual component preserves the twisted complex [5, Theorem 3.1]. We give the local algebra needed here, including its compatibility with reduction. It supplies an explicit change of basis, a stronger statement than merely saying that two dot actions induce the same homology map.

Lemma 4.2. *Let (K, d) be the ordinary crossing-cube complex of a link, with its gradings forgotten. Suppose p, p' are successive points of one actual strand, separated by one crossing. If W is any sum of fixed-arc dot actions and $W' = W + X_p + X_{p'}$, then $(K, d + W)$ and $(K, d + W')$ are isomorphic as vector spaces equipped with square-zero endomorphisms. The isomorphism preserves the reduced subcomplex at a fixed marked point.*

Proof. Use the Frobenius algebra $A = \mathbb{F}_2[X]/(X^2)$, with multiplication m and comultiplication

$$\Delta(1) = 1 \otimes X + X \otimes 1, \quad \Delta(X) = X \otimes X.$$

Direct evaluation on the basis gives

$$m\Delta = 0, \quad \Delta m = X \otimes 1 + 1 \otimes X, \quad (16)$$

$$m(X \otimes 1) = Xm = m(1 \otimes X), \quad (X \otimes 1)\Delta = \Delta X = (1 \otimes X)\Delta. \quad (17)$$

The notation on the right of the first line means the corresponding endomorphism of $A \otimes A$. Dot operators commute and square to zero; they also commute with the cube differential by (17). Thus $(d + W)^2 = 0$ for every W in the statement.

Let H be the reverse saddle at the selected crossing: it changes its cube coordinate from 1 to 0, uses the saddle opposite to the ordinary differential, and is zero in the other direction. Thus

$H^2 = 0$. Saddles at other cube coordinates commute with H ; their two occurrences in $dH + Hd$ cancel in characteristic two. At the selected crossing, (16) gives

$$dH + Hd = X_p + X_{p'} =: T.$$

Moreover $TH = 0$. If the reverse saddle merges, its two output dot actions agree and sum to zero. If it splits, the assertion follows from $(X \otimes 1 + 1 \otimes X)\Delta = 0$. Equation (17) shows that each fixed-arc dot commutes with H , including the dots appearing in W . Therefore

$$\begin{aligned} (d + W')(1 + H) + (1 + H)(d + W) &= dH + Hd + (W' + W) + W'H + HW \\ &= T + T + WH + HW + TH = 0. \end{aligned}$$

Since $(1 + H)^2 = 1$, this proves the claimed isomorphism.

If z is the reduction point, then H commutes with X_z by the same local identities. Hence $1 + H$ preserves $\text{im } X_z$, the subcomplex in which the marked resolution circle is labeled X . Its inverse does so as well. This also covers the case where z belongs to a different component from the moving dots. $\square$

Repeatedly applying the lemma moves a dot along any path in its actual component. Two unit weights meeting at one point cancel because $1 + 1 = 0$. By Lemma 4.1, the reduced cube of J_E with differential $d + X_p + X_q$ is consequently isomorphic to its ordinary reduced cube. This argument uses a sequence of local invertible maps. A general null-homotopy $T = dH + Hd$ would not, by itself, justify replacing $d + T$ by d ; the identities $H^2 = 0$ and $TH = 0$ are what make the displayed conjugation work.

4.3 Compatibility with the compressed context

The preceding argument applies initially to the crossing cube. Passing to the run complex requires compatibility of the compression with the two exterior dots. This follows from the local cobordism construction, but it is useful to record exactly which homotopy identities are needed.

Lemma 4.3. *Suppose (K, d) and (V, D_E) are related by the local Bar–Natan reductions in the context run boxes. Let L_K and L_V be the sums of the two dot actions on the distinguished smoothing arcs. Then these reductions induce a homotopy equivalence between $(K, d + L_K)$ and $(V, D_E + L_V)$ after forgetting the gradings. They also preserve the fixed reduction.*

Proof. Bar–Natan’s tangle formalism and its delooping and cancellation operations take place in the category of local dotted cobordisms [2, 3]. An exterior marked arc is retained throughout each such operation. Sliding a dot along its unchanged boundary cylinder shows that every local structural map, and every local contracting homotopy, commutes with that dot. This remains true for linear combinations and composites, and for the maps obtained by delooping a circle inside a context box. The two distinguished dots and the reduction point are outside those boxes.

More explicitly, choose the resulting maps $f : K \rightarrow V$, $g : V \rightarrow K$ and homotopies h on K , k on V . In addition to their ordinary chain-map identities they satisfy

$$fL_K = L_V f, \quad gL_V = L_K g, \quad hL_K = L_K h, \quad kL_V = L_V k.$$

Thus f and g are chain maps for the deformed differentials. Their homotopy identities remain valid, because

$$gf - 1 = dh + hd = (d + L_K)h + h(d + L_K),$$

and similarly

$$fg - 1 = D_E k + k D_E = (D_E + L_V)k + k(D_E + L_V).$$

The additional terms cancel in characteristic two. All maps commute with the reduction dot, so the same identities restrict to the reduced subcomplexes. Only finite complexes are involved; no completion or infinite-twist limit enters this argument. $\square$

Theorem 4.4 (Topological slope). *For either sign of the distinguished braid run,*

$$\boxed{\eta = \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(J_E; \mathbb{F}_2) > 0.} \quad (18)$$

The reduction point on the right is the original fixed point, and total dimension is taken over all homological and quantum degrees.

Proof. Lemma 4.3 identifies $H(V, F)$ with the homology of the reduced crossing cube deformed by the two dots. The weight-slide argument identifies that cube with the ordinary reduced cube of J_E . This proves the equality and shows that the slope is independent of the sign chosen for the distinguished run.

For strict positivity, the graded Euler characteristic of ordinary reduced integral Khovanov homology is the reduced Jones polynomial, up to the standard normalization [2]. For a nonempty link with ℓ components, its value at 1 has absolute value $2^{\ell-1}$ and is nonzero. Thus the integral reduced homology has a nonzero free summand. The universal coefficient theorem then gives nonzero reduced homology over $\mathbb{F}_2$. The diagram J_E is nonempty, so this applies even when it is a link with several components. $\square$

Remark 4.5. The dot deformation combines maps of different original bidegrees. Jaeger's formulation retains a collapsed δ grading, and forgetting that grading is sufficient for (18). The formula does not recover the individual homological or quantum degrees of $\widetilde{\text{Kh}}(J_E)$ inside the repeated part. The homological-degree profile for the original twisted closure follows instead from the separate identification of its successive complete E groups and both incident boundary maps. These are distinct assertions and use different grading information.

4.4 Computational consequences and output size

Write the context interval as $[a, c]$, let $w = c - a$, and put $m_0 = w + 2$. For $m \geq m_0$, set $\delta = m - m_0$. The two signed recurrence formulas have exactly δ inserted degrees. Combining them with Theorem 4.4 gives

$$R(m) = R(m_0) + \delta \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(J_E; \mathbb{F}_2), \quad (19)$$

where R denotes total reduced rank for either fixed choice of sign. Every inserted degree is nonzero. In particular, this is an exact linear growth statement with positive integer slope, not merely an upper bound on the possible growth.

The identity provides an independent interpretation and a possible cross-check of the slope. The algorithm need not compute $\widetilde{\text{Kh}}(J_E)$ separately: one reference computation at m_0 already contains a central boundary equal to F , so that $\eta = \dim V - 2 \text{rank } F$. The two reference shoulder Betti numbers need not equal η ; the boundary-rank extraction is essential. The original closure's component count remains necessary before using the Khovanov unknot-detection criterion [7], because shortening an exponent can change the permutation parity and the number of components.

There is also an unconditional output-size obstruction. An explicit list of nonzero degree–Betti-number pairs contains at least δ entries, and therefore requires $\Omega(\delta)$ output operations. For fixed context and growing binary-encoded m , this is exponential in the bit length of m . A compact profile avoids that cost by retaining the two fixed shoulders, their degree shifts, and one constant-coefficient interval. Its dependence on the distinguished exponent is only through binary integers of $O(\log m)$ bits. The same is true of the extra arithmetic needed to return (19). This observation does not bound the cost of computing the context data as the context grows; that cost can still be exponential. It establishes the appropriate output representation and the precise sense in which a single large exponent can be handled without expansion.

4.5 Position relative to existing tangle methods

The local twist normal form, dot calculus, and cancellation machinery are established mathematics [2, 3]. Thompson gives compact zig-zag representatives for rational-tangle Khovanov complexes [4]. Jaeger’s weight-moving theorem supplies the central algebra behind the slope identification [5]. Kotelskiy, Watson, and Zibrowius give immersed-curve descriptions of four-ended tangle invariants, gluing theorems recovering Khovanov homology, and an interpretation of the two-strand infinite-twist construction [6]. These precedents make clear why a general priority claim for twist stability would be inappropriate.

The contribution developed here is the explicit finite threshold, the extraction from one reference computation, the compact exact output, and their implementation and verification relative to the pinned ProveIt baseline. The immersed-curve viewpoint is a plausible route for studying several variable twist regions, but obtaining useful complexity bounds would require control of the context representation, local systems, and gluing calculations. The one-run theorem alone supplies no such control and does not justify a general quasi-polynomial unknot-recognition bound.

5 The compressed algorithm and its bit complexity

5.1 One finite solve and a compact exact answer

The implementation in `fastunknot.twist.tail` has a deliberately small mathematical interface. It validates the run list, selects a specified run or the largest run, and computes $[a, c]$ and m_0 . At or below the threshold it uses the ordinary macro backend. Above the threshold it changes only the selected absolute exponent to m_0 , computes the full reference homology and boundary ranks, and applies theorem 3.2.

The returned degree profile consists of a dictionary of exceptional degree–dimension pairs and at most one interval

$$(\ell, u, \eta), \quad \text{meaning } \beta_h = \eta \text{ for every integer } \ell \leq h \leq u.$$

The exceptional degrees are disjoint from that interval. Endpoints are inclusive. A single-degree query uses these two components directly. The total rank is the sum of the point coefficients plus $(u - \ell + 1)\eta$ for each interval. An explicit expansion routine is available, but has a caller-visible limit on the number of emitted terms.

```

validate the original braid and retain its component count
choose the distinguished run, with sign epsilon and length m
compute context endpoints a, c and threshold m0 = c - a + 2
if m <= m0:
    return complete ordinary macro homology as a point profile
reference = the same signed runs with selected length m0
solve the complete reference complex
hc = c + 1 if epsilon > 0 else a - 2
M = reference.chain_dimensions[hc]
r = reference.boundary_ranks[hc]
eta = M - 2*r
shift the appropriate reference boundary region by m - m0
insert the proved constant interval, if it is nonempty
return exact rank, compact profile, original components,
    reference output and recurrence certificate

```

The certificate records the threshold, degree endpoints, selected position, sign, reference dimensions, central boundary rank, slope, and finite solve. It is a transparent computational witness. The implementation does not claim that arbitrary certificate JSON is a formal proof of the topological normal-form theorem. The independent audit reconstructs the recurrence without calling the production recurrence routine.

5.2 A precise parameterized guarantee

Theorem 5.1 (Cost in the context and exponent). *Let a fixed-context family be as in theorem 3.2. Let $\mathcal{N}_0$ be its reference chain dimension. Its exact compact reduced homological profile can be computed using one ordinary reference solve, with safe time bound*

$$\text{poly}(b, W) \mathcal{N}_0^3 + \text{poly}(W, \log(m+1), \log(\mathcal{N}_0+1)). \quad (20)$$

The exponent-dependent part of the output has polynomial bit size in $W + \log(m+1) + \log(\mathcal{N}_0+1)$. For original one-component closures with at least one run, this gives the coarser uniform bound

$$2^{O(W+1)} \text{poly}(\log(m+1)). \quad (21)$$

Fixed resource caps can interrupt the reference solve; in that event the bounded implementation makes no exact-rank claim.

Proof. The reference has $2W + 2$ expanded crossings. There are at most W other nonempty runs, so $t \leq W + 1$. Each reference state can be constructed and its local Frobenius maps evaluated in a polynomial in b, W , times the number of labels processed. Assembling the reference differential and applying ordinary elimination is therefore bounded by the first term of (20). This intentionally loose cubic bound suffices regardless of the practical bitset improvements.

The reference homological support has at most $2W + 3$ possible degrees. Splitting, shifting, and summing these entries takes polynomially many integer operations. All reference coefficients and boundary ranks are at most $\mathcal{N}_0$, and the final rank is at most $\mathcal{N}_0 + m\mathcal{N}_0$. Degree shifts use integers of $O(\log(m + W + 2))$ bits. Standard integer arithmetic proves the second term and the output claim.

For the uniform knot bound, every generator occurs in the original braid. At most one generator index is supplied only by the selected run, so $b - 2 \leq W$. A support of size s has at most $b + s$

resolution circles: changing one I into E changes component count by at most one. Hence, even without the sharper first-use bound from report 11,

$$\mathcal{N}_0 \leq 2^{b-1} \prod_j (1 + 2m_j^{(0)}) \leq 2^{b-1} 3^{2W+2} = 2^{O(W)}.$$

Here $1 + 2u \leq 3^u$ for integers $u \geq 1$, and $m_j^{(0)}$ are the reference lengths. Substituting this into (20) proves (21). The reference may be a link; the argument only uses occurrence of the original generator indices, which shortening preserves. $\square$

For fixed context the finite solve is a constant with respect to m , and the remaining bit complexity is polynomial in $\log m$. This is the encoding improvement over expanding all $m + 1$ local terms. In particular $W = O((\log n)^2)$ gives $n^{O(\log n)}$, and $W = O((\log \lambda)^2)$ gives $\lambda^{O(\log \lambda)}$. More generally, if $W = O((\log n)^k)$ for a fixed k , then (21) is a quasi-polynomial homology bound in the expanded size n . If the same relation is imposed with the encoded size λ , it is a quasi-polynomial bound in λ . These are promises on the supplied context, not theorems that arbitrary diagrams admit such a representation.

Several comparable huge runs do not satisfy the fixed-small-context condition. For example, if two lengths are both of order m , choosing one leaves W of order m . Applying the theorem once does not remove that dependence. Any run satisfying $m \geq W + 2$ has more than half the total crossings, so there can be at most one. Choosing the largest run therefore finds this regime whenever it is present. This fact is not a multivariable compression theorem.

5.3 What can and cannot be inferred from a shortened reference

The original strand permutation is indispensable. A two-strand run of length 101 is a knot, while its reference length $m_0 = 2$ is a two-component link. Nevertheless the marked reference homology is the correct algebraic input to the recurrence. Using the reference component count in a recognition wrapper would be a bug.

The original crossing counts are also indispensable for a grading adapter. The existing macro-to-scanner conversion uses

$$h_{\text{cube}} = n_+^{\text{original}} - h_{\text{macro}}.$$

Substituting the shortened positive crossing count would shift the reported homology incorrectly. The delivered raw interface retains the macro grading explicitly. The existing one-component characteristic-two adapter can double reduced dimensions to obtain its unreduced output, but this knot-specific convention should not be silently transferred to general marked links.

The flag `check_d2=True` checks the actual finite reference differential. It does not construct and multiply an enormous original matrix. The output says which finite complex was checked. General correctness for all larger exponents rests on the proof of the recurrence.

6 Structural certificates directly from braid runs

An exact algebraic improvement is valuable even when recognition can be settled more cheaply. The audit makes this separation particularly sharp for the dominant-run family. We first specialize the existing structural bounds to braid counts, then prove why a proposed signature shortcut would add no cases.

6.1 The signed path multigraph

Assume the original closure is one component. Let p_i, n_i be the numbers of positive and negative occurrences of generator i . Define

$$P = \sum_i p_i, \quad N_- = \sum_i n_i, \quad n = P + N_-, \quad e = P - N_-,$$

$$u_+ = |\{i : p_i > 0\}|, \quad u_- = |\{i : n_i > 0\}|, \quad d = n - b + 1.$$

The b oriented Seifert disks of a braid form a path multigraph, with all crossings of σ_i joining vertices $i, i + 1$. The positive and negative subgraphs have $b - u_+$ and $b - u_-$ connected components, respectively, including isolated vertices. Specializing the Kawamura–Lobb bound and the homogeneous-diagram criterion [9, 10, 11] gives

$$e + b - 2u_+ - 1 \leq s(K) \leq e - b + 2u_- + 1, \quad (22)$$

in the usual Artin crossing-sign convention. The canonical surface has genus $d/2$, and the homogeneity defect is

$$u_+ + u_- - b + 1 = |\{i : p_i > 0 \text{ and } n_i > 0\}|. \quad (23)$$

The equality follows because every generator index occurs in at least one sign.

Thus $d = 0$ certifies the unknot. If the defect is zero, the homogeneous diagram realizes the knot genus; positive d certifies knottedness. An interval in (22) excluding zero also certifies knottedness. An interval containing zero alone does not certify triviality, even if it consists of the single value zero.

Proposition 6.1 (Encoded structural evaluation). *All these quantities and the original one-component condition can be computed in polynomial time in the run-encoded input size λ , without expanding exponents.*

Proof. Accumulate signed occurrence counts and exponent sums using integer arithmetic. Apply the adjacent transposition for each odd exponent, keeping only touched positions in a sparse permutation. A position not touched is fixed; add those fixed cycles to the cycles in the sparse permutation. If some generator index is absent, reject the knot-only interface without allocating an array of length b . Otherwise, for a knot candidate $b - 1 \leq t$, so all count and cycle structures have polynomial size in λ . The displayed formulas then require only additions, comparisons, and small-integer multiplications on numbers of $O(\lambda)$ bits. $\square$

The module `braid_profile.py` implements this specialization. One repository sign convention needs explicit documentation. For inputs built by `Diagram.from_braid`, the production Seifert evaluator assigns the opposite sign to the usual positive Artin letter. Consequently the public fields matching existing Seifert output use $-e$ and interval $[-U, -L]$, where $[L, U]$ is (22). The module also returns `artin_exponent_sum` and `artin_rasmussen_interval`. This sign conversion leaves every recognition conclusion unchanged. Exhaustive comparisons check the complete existing field dictionary, not just the final verdict.

6.2 A sound signature idea that is dominated

The proposed shortcut was to retain many same-sign bands in the canonical Seifert surface and use a large definite subspace of its symmetrized form. The mathematical argument is valid, including when selected occurrences are nonconsecutive in the original word.

Lemma 6.2 (Retained bands). *Retain all b Seifert disks and $p \geq 1$ same-sign occurrences of one generator. The retained surface contributes a definite subspace of dimension $p - 1$ to the symmetrized Seifert form of the full surface. For a fixed sign, nonadjacent generator indices contribute a direct sum of such subspaces.*

Proof. The retained surface is the standard $T(2, p)$ band surface, together with unused disks. Its symmetrized form is definite of rank $p - 1$, with sign determined by the crossing convention [12]. Deleting other bands does not alter the relative order of the retained same-generator bands. The full surface is obtained by attaching only one-handles to this retained surface F_0 . Hence $H_2(F, F_0) = 0$, and the relative homology sequence gives an injection $H_1(F_0) \rightarrow H_1(F)$. The Seifert pairing restricts to the pairing of F_0 . For nonadjacent indices the retained bands use disjoint strand pairs and form a split union after deleting all other bands; their forms sum directly. Retaining all disks is relevant to the injection argument: arbitrary subsurfaces need not have injective first homology. $\square$

For one sign let $w_i = \max(p_i - 1, 0)$, using n_i for the other sign. A maximum-weight independent set of the index path chooses nonadjacent generators with total weight R . It is computed by

$$B_j = \max(B_{j-1}, B_{j-2} + w_j).$$

A definite R -dimensional subspace in a symmetric form of dimension d gives the sufficient signature inequality

$$|\sigma(K)| \geq 2R - d \quad \text{when the right side is positive.} \quad (24)$$

Indeed, at least R eigenvalues have the selected sign and at most $d - R$ can have the opposite sign.

Proposition 6.3 (Domination). *Every braid knot certified nontrivial by (24) is already certified by (22). Signature packing adds no instances to the existing structural stage.*

Proof. For positive bands,

$$R \leq \sum_i \max(p_i - 1, 0) = P - u_+.$$

The lower endpoint of (22) equals

$$2(P - u_+) - d.$$

It is at least $2R - d$. Therefore a positive signature packing bound forces a positive Rasmussen lower bound. The negative version gives an upper bound at most $-[2(N_- - u_-) - d]$, and the same conclusion follows when negative packing succeeds. $\square$

The diagnostic signature routine and witness checker are retained for reproducibility, but the recognition path uses the cheaper dominating counts directly. This rejected optimization is a useful result of the audit: a correct new certificate is not automatically a stronger one.

Corollary 6.4 (Every dominant-run knot is already decided). *If a one-component b -braid has a selected run of magnitude $m \geq W + 2$, then its existing structural Rasmussen interval excludes zero. In particular it is knotted.*

Proof. For a positive selected run,

$$P - u_+ \geq m - 1, \quad 2(P - u_+) - d \geq m - W + b - 3 \geq b - 1 > 0.$$

Here $d = m + W - b + 1$, and the presence of a run implies $b \geq 2$. For a negative selected run the upper endpoint is at most the negative of the same positive quantity. $\square$

This corollary fixes the interpretation of the performance results. The recurrence computes much more homological information at much lower cost on its long-run families, but it does not create a new class of previously undecided knots in that range. The direct run-based structural interface, on the other hand, makes the already known decision available without first paying the expanded-input cost.

7 Degree-streamed exact homology

7.1 Enumeration in the correct signed degree

The streamed backend evaluates exactly the same macro complex as `twist.core`. It changes storage and evaluation order. Set $\nu = \sum_{e_j < 0} m_j$ and replace each local coordinate by

$$\alpha_j = \begin{cases} k_j, & e_j > 0, \\ m_j - k_j, & e_j < 0. \end{cases}$$

Then every differential increments one α_j , and the total degree is $h = \sum_j \alpha_j - \nu$. For negative runs the support condition is $\alpha_j \neq m_j$, not $\alpha_j \neq 0$. This is a necessary detail for mixed-sign inputs.

To enumerate degree h , solve the bounded composition $\sum_j \alpha_j = h + \nu$, with $0 \leq \alpha_j \leq m_j$. If the remaining sum at position j is R , and $M_{j+1} = \sum_{i>j} m_i$, visit only

$$\max(0, R - M_{j+1}) \leq \alpha_j \leq \min(m_j, R). \quad (25)$$

Every visited prefix extends to a solution, since the remaining interval coordinates realize every integer from zero through their total capacity. Conversely every valid state obeys these bounds. The iterative depth-first implementation therefore emits each state once, uses $O(t)$ generator memory, and avoids rescanning the entire Cartesian product in each degree.

If s_h is the number of states in degree h , there are at most s_h feasible prefixes at each depth. The total enumeration cost is $O(tS)$, because $\sum_h s_h = S$ and $n + 1 \leq S = \prod_j (m_j + 1)$ for nonempty run lists. The empty list is handled separately.

7.2 Two adjacent indexes and generated columns

At degree h , keep the source state index and construct the target index for degree $h + 1$. An index entry contains the coordinate tuple, support mask, basis offset, and basis dimension. These two indexes determine every local edge and its destination offset. Generate a differential column, immediately reduce it against a dictionary of pivot columns, and discard it if dependent. Only pivots are retained for rank computation.

When r_h is complete, (7) finalizes degree h . Release its source index and pivot dictionary, retain r_h as a scalar, and advance. By induction each local map occurs exactly once in the correct matrix. The degree-wise basis order may differ from the ordinary backend for negative runs, but this only permutes matrix bases. Exact elimination therefore gives identical chain dimensions, boundary ranks, and homology.

Geometry is obtained from the pinned union-find construction. The geometry cache has an entry cap and a cap on total cached vertex occurrences. The map cache has an entry cap and a cap on descriptor integer slots. Both are least-recently-used caches; an evicted support is not retained in an unbounded history set.

Local maps use compact circle-index descriptors. A merge relabels circles and sends two X 's on the same output circle to zero. A split evaluates $\Delta(1)$ or $\Delta(X)$. A dot map stores two circle indices, cancelling coincident dots and killing multiplication on the marked X -circle. This avoids a cached table with one entry for each of 2^{c-1} labelings. The current source state's outgoing descriptors also have an explicit active-slot cap.

7.3 A storage bound that includes auxiliary data

Let $V_g = b(t+1)$ be the geometry grid size, let $L_s = \max_h(s_h + s_{h+1})$, and let C_g, C_m, A_m be the actual peak cached geometry vertex occurrences, cached map slots, and active map slots. On a word model large enough for coordinates and offsets, the ordinary streamed computation uses

$$O\left((t+1)L_s + (n+1) + C_g + C_m + A_m + V_g + t + \max_h(r_h + 4)(1 + \lceil d_{h+1}/w_{\text{bit}} \rceil)\right) \quad (26)$$

words, where w_{bit} is the bitset limb size. The four-vector allowance covers a yielded column, its evolving reduced value, a shifted unit bit, and an allocated XOR result. The $O(n+1)$ term is the retained full output and scalar degree statistics. Omitting it would wrongly suggest constant total memory for a long two-strand explicit-profile computation.

For binary coordinates, replace scalar word counts by their bit lengths. For example the two indexes have the safe bound

$$O(L_s[t \log(n+2) + t + \log(\mathcal{N}+1)])$$

bits, apart from container overhead. Geometry and descriptor indices use $O(\log(V_g+1))$ bits, and support keys use $O(t)$ bits. Finite transient geometries and descriptors are included as well as cached entries.

Without checking mode, the conservative algebraic bit envelope is

$$\max_h(r_h + 4)d_{h+1}.$$

With `check_d2=True`, retain two adjacent matrices and compose them column by column; the old source state index is unnecessary. Thus there are still only two state indexes. A safe algebraic envelope in checking mode is

$$\max_h \{d_{h-1}d_h + d_h d_{h+1} + \max((r_h + 4)d_{h+1}, 4d_h + 3d_{h+1})\}. \quad (27)$$

Pivots are released before the composition phase. Matrix and pivot reference counts are capped separately, so a large list of zero columns does not evade accounting through its zero bit payload.

The ordinary backend retains all state indexes and all matrices, with bit envelope $Q = \sum_h d_h d_{h+1}$. Replacing that sum by adjacent degree costs can save substantial storage, but the bounds are mathematical payload bounds, not measured Python heap or process-RSS ratios. The experiments measure newly traced Python allocations separately.

7.4 Time, budgets, and exact early decisions

Let G count actual geometry constructions, M_{map} map compilations, and E_h emitted local differential terms in degree h . Safe bounds are

$$G \leq (t + 2)S, \quad M_{\text{map}} \leq tS, \quad E_h \leq 2td_h.$$

Neighbor-tuple construction can cost $O(t^2S)$. The particular union-find implementation has a safe elementary $O(V_g^2)$ construction bound; the analysis does not assume a stronger bound its code has not established. In addition to topology and label evaluation, column assembly and elimination cost at most

$$O\left(\sum_h [E_h + d_h(r_h + 1)] (1 + \lceil d_{h+1}/w_{\text{bit}} \rceil)\right)$$

word operations. Each pivot XOR lowers the leading set bit, so a column uses at most r_h pivot XORs. Optional differential-square checking adds its actual column-composition work. Bounded caches can repeat geometry and map work. These facts explain why a memory improvement need not be a time improvement.

The budget distinguishes total macro states from live states, total enumerated basis from one layer's basis, and live algebraic payload from cache storage. Product bounds are checked before expanding states. Huge exponents and huge strand counts are rejected by arithmetic guards before large allocations or decimal diagnostic formatting. The deadlines are cooperative, and the units are structural counts, not hard process-memory reservations.

After degree h is finalized, define

$$R_{\leq h} = \sum_{i \leq h} (d_i - r_{i-1} - r_i).$$

Every summand is an actual nonnegative homology dimension. For an original one-component closure, $R_{\leq h} > 1$ therefore proves **KNOTTED**. Later degrees cannot cancel homology already computed. An early result leaves `reduced_rank` unset and reports `reduced_rank_lower_bound=R`. Its completeness field is `homology_complete=False`. If checking mode is enabled, the checked prefix is reported and the full-complex check flag remains false. **UNKNOT** requires a complete rank-one calculation.

For a positive two-strand run of odd magnitude at least three, four states suffice for an early nontriviality result; three suffice for the negative run. These are transparent raw-backend demonstrations. Such examples are already easy for the structural front end and do not demonstrate acceleration of a filter-resistant recognition workload.

8 Validation and measured performance

8.1 Correctness checks and what they establish

The complete integrated suite passed **254 test methods**, with zero failures or errors, in 36.700 seconds on Python 3.12.14. Of these, 209 are inherited methods and 45 belong to the four new test modules. The execution log and a machine-readable summary are included. Many methods contain whole finite enumerations, so a test-method count is not the number of individual algebraic comparisons.

Check	Coverage
Tail module, 11 methods	400 seeded complete-profile comparisons with the macro backend; 80 comparisons with the independently assembled crossing cube; both signs, arbitrary selected positions, threshold endpoints, links, original parity, resource failure, and enormous exponents.
Streaming module, 17 methods	468 exhaustive signed words against the independent cube, plus six dedicated examples; all compact map types, degree and boundary ranks, cache caps, differential-square checks, live-state counts, and early decision semantics.
Structural module, 8 methods	3,026 exhaustive checked source-versus-PD summaries; 240 higher-strand seeded pipeline comparisons; signature domination, path optimization, invalid inputs, and 100,001-bit exponent arithmetic.
Continuation interface, 9 methods	All three exact backends, structural-first recognition, complete link homology, original component checks, resource UNKNOWN, stdin, configuration errors, and exit codes.
Separate recurrence audit	104 contexts and 208 signed families; 832 macro and 220 independent crossing-cube comparisons, without calling the tail implementation.
Separate slope audit	500 comparisons of the central-matrix slope with ordinary reduced homology of the extracted turnback-context complex.

The exhaustive word sets include equivalent words, repeated knot types, links, and empty words; they are not catalogues of distinct knots. The independent crossing cube does not use the run recurrence or its local compression, which makes it a meaningful implementation cross-check. It still shares the characteristic-two Frobenius theory and some representation conventions. Neither repeated implementations nor finite tests are substitutes for the mathematical proof.

Two large-encoding checks illustrate the exact scope. For both signs, a two-strand exponent $m = 2^{100000} + 1$, having 100,001 bits, is handled from a four-element reference basis. The result has two exceptional degrees and one interval, and its exact rank is m . For

$$\sigma_1^3 \sigma_2^{-3} \sigma_3^{10^{100}+1} \quad \text{on four strands,}$$

the returned rank is $9(10^{100} + 1)$, the slope is 9, and the reference basis has fewer than 1,000 elements. These checks exercise big-integer arithmetic and avoid exponent expansion; they do not certify that a reference of arbitrary context size will fit the resource budget.

8.2 Paired long-run homology measurements

The recurrence benchmark uses seven A/B/A rounds per case, a separate seven-pair A/A control, and fresh computation caches. A is the ordinary macro backend; B is the exact recurrence. A returns an explicit degree dictionary and B an exact compressed profile. Their complete homological information agrees. Expansion of B's compact interval is excluded from its timed task, because compact output is the interface whose complexity is being measured.

The benchmark contains three families at selected lengths 17, 65, 257, 1025, plus a below-threshold no-compression control. There is no braid simplification or earlier recognition filter in these timings. The three largest-length results are:

Family, $m = 1025$	$\mathcal{N}(m)$	$\mathcal{N}_0$	A (ms)	B (ms)	Paired A/B
Two strands	1,027	4	3.639	0.084	39.42
Figure-eight context	15,393	93	44.000	0.637	69.12
Mixed four-strand link	18,486	126	37.998	0.931	44.05

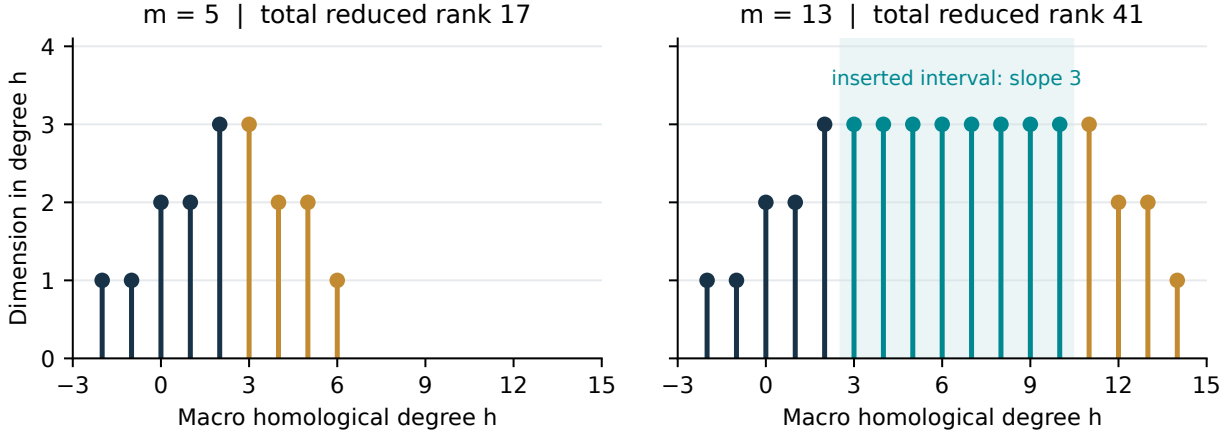


Figure 1: Exact reduced homological profiles of $\sigma_1^m \sigma_2^{-1} \sigma_1 \sigma_2^{-1}$. The dark lower region is fixed, the gold upper region translates by $m - 5$, and the teal interval contributes rank 3 in each inserted degree. Both panels are generated from exact output; the recurrence is proved independently of the plotted examples.

Durations are medians, with A averaged over the two surrounding calls in each round. The final column is the median of the paired round ratios, not the quotient of the separately displayed medians. The mixed four-strand input is $\sigma_1^m \sigma_2^{-1} \sigma_3 \sigma_2^{-1}$; its measured odd- m closure is a link, and the benchmark computes its marked reduced homology.

The no-compression control had median A/B 1.021, with individual ratios from 0.723 to 1.355. The separate A/A control had median second/first 1.064. Small durations are visibly noisy. The algebraic count reduction, the exact equality checks, and the asymptotic theorem are stronger evidence than a single microbenchmark ratio. These are standalone full-homology gains. By corollary 6.4, the knot families in the proved long-run range are already structurally decided by the baseline.

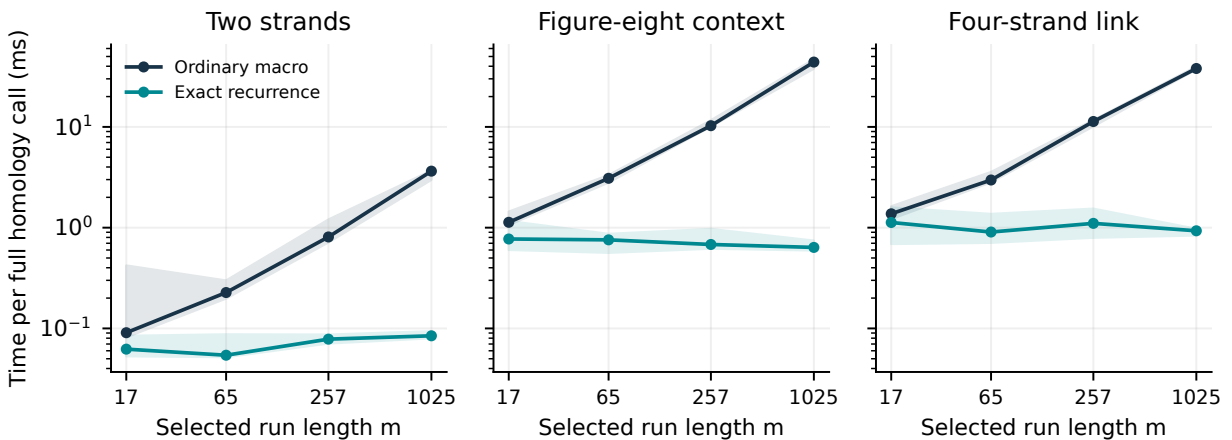


Figure 2: Median standalone homology times at four run lengths. Shading shows the interquartile ranges of the retained paired samples. Both axes are logarithmic. The finite reference is unchanged within each family; the measurements still show ordinary host and timing variability.

8.3 Streaming: memory savings with an observed time cost

The streaming experiment uses eight cases and seven randomized rounds, with five fresh calls per arm per round. Each round includes an identical original-backend control. Full ranks, homological profiles, chain dimensions, and boundary ranks are compared. Differential-square checking occurs before timing. Memory tracing occurs in separate fresh runs, so its instrumentation does not contaminate the duration measurements.

Across these cases, the paired original/streamed time ratios range from 0.265 to 0.698: complete streamed homology is slower. The original/streamed peak traced-allocation ratios range from 1.028 to 3.064: peak allocation is smaller. A/A time ratios range from 0.849 to 1.198. The allocation measurements are newly traced Python allocations from `tracemalloc`, not process RSS or a hardware memory lower bound.

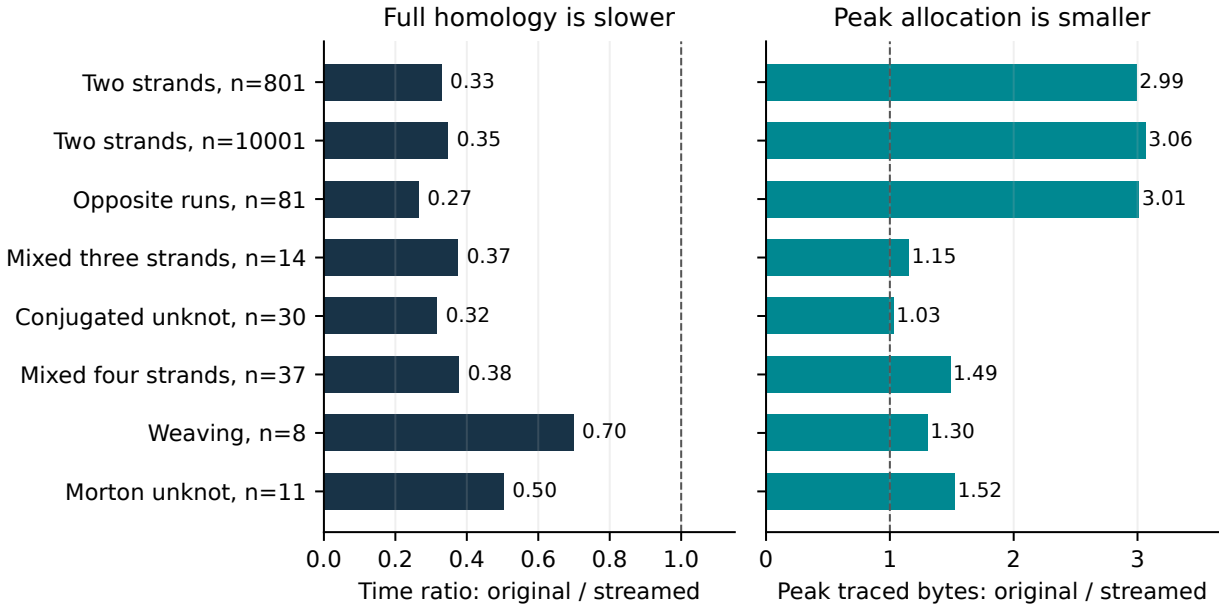


Figure 3: The full-homology time–space tradeoff across all eight streamed benchmark cases. A ratio below one on the left means the streamed implementation is slower. A ratio above one on the right means it uses fewer peak traced bytes. The dashed line is equality.

The repository’s actual Morton input is an informative complete unknot example:

$$[-3, -3, 2, -3, 2, 1, 1, 1, -2, 1, -2] \quad (b = 4).$$

It has 768 macro states and 2,799 reduced basis elements. In degrees $-5, \dots, 6$, its chain dimensions and outgoing ranks are

h	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
d_h	8	48	138	270	417	518	510	414	278	142	48	8
r_h	8	40	98	172	245	272	238	176	102	40	8	0

so (7) gives one class in degree zero and no others. The streamed run retains at most 308 state entries and 272 pivots. The dense full-matrix sum is $Q = 1,009,952$ bits; the ordinary streamed live envelope is 140,760 bits. These are two structural bounds, not measured byte values.

The measured Morton medians are 27.360 ms for the original and 50.385 ms for streaming. Peak traced allocations are 1,441,224 and 946,172 bytes, respectively, a 1.523-fold reduction. Thus the implementation improves memory use on an actual full-rank-one calculation but does not justify replacing the faster default on time grounds.

For σ_1^{10001} , full original and streamed computations take 37.742 and 100.604 ms, while streamed early recognition takes 0.118 ms and visits four states. The four-strand mixed case of 37 crossings visits 192 of 2,048 states before an early lower bound exceeds one. These demonstrate the correctness and possible value of an early backend endpoint. They do not measure its value after the existing structural and modular filters.

8.4 Separating encoding savings from recognition savings

The structural experiment uses

$$\gamma_m = \sigma_1^m \sigma_2^{-(2m-1)} \sigma_3^m, \quad b = 4, \quad m \geq 3 \text{ odd}.$$

These are one-component homogeneous braids with exponent sum 1, zero Rasmussen interval, and positive canonical genus. The homogeneous-genus criterion already proves nontriviality. The coarse exponent-sum obstruction alone does not do so.

One experiment times a direct RLE certificate against expansion of the word, checked PD construction, and the existing Seifert certificate. A second uses fresh prebuilt diagrams and changes only the opt-in `use_braid_profile` argument to the recognizer; construction is outside that second timer.

m	Crossings	Expanded construction / RLE	Profile off / on	Pipeline A/A
17	67	107.71	2.76	0.964
129	515	999.73	4.49	1.168
1025	4099	6645.13	4.03	1.039
4097	16387	36063.64	3.48	0.884

All ratios are medians of paired rounds. The large construction-inclusive ratios measure the explicit encoding difference: three integer runs versus thousands of crossings and PD incidences. They are not general recognition speedups. The prebuilt-diagram experiment shows an additional constant-factor improvement on these selected structural cases. All decisions and all compared Seifert metadata agree. The Morton structural-only control remains **INCONCLUSIVE** in both implementations; its full recognizer is excluded from that experiment.

The first shorter batches showed substantial timer noise. Both pilot files are retained, and the final script uses fixed size-based larger batches rather than selecting favorable samples. The reported A/A controls still require approximate interpretation. The polynomial encoded-input theorem is independent of those timings.

9 What still obstructs a general quasi-polynomial bound

9.1 Finite matching types do not bound multiplicity

The macro state space separates geometry from algebraic copies. Even a very small collection of boundary matchings can occur with exponentially large multiplicities. Streaming changes how those

copies are visited; it does not eliminate the copies. The one-run recurrence does eliminate a repeated algebraic region, but only after a fixed finite context has been solved.

The three-strand obstruction in Kelomäki–Schütz [8] is particularly decisive against a universal width-only argument. Some constant-strand families have exponentially growing total Betti number, so any algorithm explicitly retaining a basis must expand exponentially. Their specialized polynomial computation shows why this is a representation obstruction rather than an impossibility theorem for compact output. The same lesson applies to scalar transfer matrices: their coefficients can encode very large counts, but a transfer formula for the Euler characteristic does not automatically determine homology ranks.

Likewise, the restriction $t = O(\log n)$ in (2) is substantive. It cannot be removed by calling the supplied run count a width parameter. A general diagram may have $t = \Theta(n)$, and the existence of some small topological decomposition does not imply a polynomial procedure to find it with short, verifiable moves.

9.2 Why several long variables need more than interpolation

A plausible next direction is to allow several large exponents at once. Even the simplest example rules out one overly strong formulation.

Proposition 9.1 (No single eventual polynomial in all twist lengths). *There is no polynomial $p(m, \ell)$ agreeing, for all sufficiently large m, ℓ with $m - \ell$ odd, with the total reduced characteristic-two Khovanov rank of the closed braid $\sigma_1^m \sigma_1^{-\ell}$.*

Proof. The braid cancels to $\sigma_1^{m-\ell}$, a two-strand torus knot when $m - \ell$ is odd. Its reduced rank is $|m - \ell|$. This last fact also follows directly from the one-run normal form: there is one class in degree zero, none in degree of magnitude one, and one in each subsequent degree through magnitude $|m - \ell|$.

If p existed, fix any sufficiently large integer ℓ . For infinitely many admissible $m > \ell$, $p(m, \ell) = m - \ell$. A one-variable polynomial with infinitely many zeros is zero, so $p(x, \ell) = x - \ell$ identically in x . This holds for infinitely many ℓ , hence $p(x, y) = x - y$ as a two-variable polynomial. It contradicts the required value $\ell - m$ for admissible $\ell > m$. $\square$

The cancellation in this example is cheap and should be performed first in a practical recognizer. Its role is conceptual: comparable exponents can produce different asymptotic chambers separated by equalities of lengths. A multivariable theorem might use piecewise polynomial, quasi-polynomial, or rational generating descriptions with explicit chamber conditions. A few numerical values cannot select the correct description or prove its validity throughout a large orthant.

9.3 A sufficient representation target

The useful target is a compact algebraic model closed under the operations the algorithm actually performs. For a tangle fragment it must include boundary data, grading shifts, multiplicities, differentials, and the maps used in subsequent gluing. A list of dimensions is insufficient.

Proposition 9.2 (Conditional complexity contract). *Suppose every n -crossing input admits a constructible decomposition with $\text{poly}(n)$ composition steps. Assume:*

1. *each intermediate fragment has a faithful algebraic description of at most $S(n)$ bits, including all multiplicities and map data;*
2. *each composition, reduction, and final rank-one decision can be performed in $\text{poly}(n, S(n))$ bit operations while preserving a description of that size;*
3. *the construction and the correctness of every representation change have the same total complexity bound.*

If $S(n) = 2^{O((\log n)^2)}$, this gives a complete $n^{O(\log n)}$ -time recognition algorithm.

Proof. There are polynomially many steps, each of polynomial degree independent of n in n and $S(n)$. Their sum, including the construction and final decision, remains $2^{O((\log n)^2)}$. Faithfulness makes the final rank-one decision equivalent to (5). $\square$

This proposition is a contract for future theorems, not a proof that such descriptions exist. In particular the first assumption forbids hiding large basis multiplicity inside a supposedly small number of geometric types, and the second forbids an implicit exponential expansion during composition. The third distinguishes finding a useful decomposition from merely asserting its existence.

The present recurrence satisfies this kind of contract for one finite context followed by a homogeneous repeated chain region. It gives a concrete example of how much information can be compressed while still recovering exact ranks. Extending that closure property to interacting regions is the central algebraic problem.

10 Further research questions and proposed work programme

The following topics are ordered by their connection to the actual remaining bottlenecks. Each has a concrete mathematical or experimental completion criterion. They are proposals, not claimed consequences of the current implementation.

10.1 Reduce the effective context amplitude

Research question 10.1. Can the raw crossing span W be replaced by the degree span of a smaller context model that retains both dot actions and the distinguished saddle?

The proof of theorem 3.2 uses a common support interval for the I and E context sectors and compatibility with their maps. Suppose an explicitly constructed equivalence of this two-sector context has degree-preserving forward and backward maps, with contracting homotopies, all compatible with both exterior dot actions and the distinguished saddle. If both resulting sector complexes are supported in $[a', c']$, the same proof gives threshold $c' - a' + 2$. Compatibility is required at the level of maps and homotopies. An ordinary homotopy equivalence of the underlying complexes alone does not suffice; transporting operators through an arbitrary contraction can introduce additional terms.

A first implementation should perform local contractions in the open tangle category and transport the saddle, dot actions, and contracting homotopies. It should output an equivalence witness and

measure both the reduced amplitude and the resulting matrix size. A useful negative outcome would be a family where amplitude is small but the equivariant context dimension still grows exponentially. That would isolate the next barrier more accurately than crossing count alone.

10.2 Two interacting variable regions before arbitrary many

Research question 10.2. For two distinguished runs, can their simultaneous long region be described by finitely many exact face and chamber models whose evaluation is polynomial in the two exponent bit lengths?

The one-run interior is controlled by $D_E + L$. With two selected turnbacks, there are two commuting local dot sums and several I/E sectors. The total homological degree combines both local coordinates, so the number of copies along a diagonal changes near the boundary of the exponent rectangle. A faithful model must retain those incidences, not simply add two independently computed slopes.

Begin with adjacent and nonadjacent generators, both relative signs, all original component parities, and the cancellation family in [proposition 9.1](#). Require an exact symbolic formula whose chamber conditions are proved, then compare complete small profiles with the crossing cube. Only after the two-variable case is understood should one study an s -variable face system with up to 2^s sectors. Even that sector count would not bound internal module complexity.

10.3 Restore quantum information

Research question 10.3. What is the exact bigraded refinement of the one-reference recurrence, including the shifts of both boundary regions and the repeated interval?

The ungraded deformation $D_E + L$ combines maps with different original bidegrees. The slope identity gives total dimension and does not place the smoothing's original bigraded groups in the long complex. A correct refinement must carry the quantum shifts in the integer-twist normal form before flattening the central groups. One possible output is a rational generating expression in homological and quantum variables, with finite boundary corrections. Its validity and output size have to be proved. The completion criterion is agreement with an independently bigraded cube on both signs and mixed contexts, not just recovery of the Jones Euler polynomial.

10.4 Odd characteristics and integral torsion

Research question 10.4. Can a signed periodic central model replace F over fields of odd characteristic and over the integers?

The current square-zero calculation and weight cancellation use characteristic two. Over the integers, twist differentials alternate dot sums and differences; signs also interact with tensor degrees. A period-two central model is a natural object to examine, but using the present $D_E + L$ unchanged is invalid. An integral algorithm additionally needs Smith-form or module data, with the bit lengths of torsion invariants included in its complexity. The first result should be a fully signed local and contextual theorem, followed by a recurrence for isomorphism types or a precisely weaker rank statement.

10.5 Decision certificates smaller than full homology

Research question 10.5. Can a rank-greater-than-one witness be represented and verified compactly on inputs that evade the structural and modular filters?

There is a concrete algebraic witness to seek. In a chain complex with total differential d , find two cycles z_1, z_2 and two linear functionals ℓ_1, ℓ_2 satisfying

$$dz_j = 0, \quad \ell_i d = 0, \quad \ell_i(z_j) = \delta_{ij}.$$

Then the two homology classes are independent: any boundary relation between them is annihilated by both functionals. This elementary criterion gives a verifiable target. An explicit witness can be exponentially long, so the research problem is a representation that supports these products without expansion. The streamed lower bound already proves a decision but does not yet produce such a portable compressed cycle–cocycle witness.

Unknot certificates need a different positive mechanism, such as a small contraction to the one-generator reduced complex or a topological simplification certificate. A method that quickly produces two surviving classes on many knots still needs a complete, costed endpoint on hard unknots.

10.6 Use tangle invariants with explicit algebraic size accounting

Research question 10.6. Can immersed-curve or related four-ended tangle descriptions support exact repeated gluing with compact multiplicities and controlled local systems?

The framework of Kotelskiy–Watson–Zibrowius is a serious source of structure [6]. The relevant complexity parameters would include curve segments, intersections under twisting, local-system ranks and entries, and the cost of cancelling the resulting Floer-type complex. A picture with few curves may still encode many intersections or a large local system. The first milestone should translate one delivered long-run example into that representation and recover its threshold, slope, and boundary corrections with explicit cost accounting.

10.7 Recover speed while preserving the streamed memory bound

Research question 10.7. Which bounded low-level changes recover the streamed backend’s observed time loss without restoring unbounded tables?

Candidates include a compiled exact bitset kernel, small-label tables admitted only under an explicit slot cap, reused neighboring state keys, and a more efficient geometry construction. The benchmark currently shows a real time cost, so a default switch would be premature. Any follow-up should report complete output equality, paired A/B and A/A timings, total versus live state counts, map-compilation counts, and separate traced-allocation measurements on the same inputs.

A related question is whether the sweep should start at the other homological end. Reversing the sweep requires the correct transposed local maps and grading conversion. A preflight estimate may choose an end, but an estimate of where homology is likely cannot substitute for the exact incident ranks required by an early decision.

10.8 Benchmark the unresolved part of the pipeline

Research question 10.8. Which checked inputs actually reach the expensive endpoint after all current inexpensive certificates, and what algebraic feature predicts their cost?

Record the exit stage of every case, rather than timing all backends on examples already decided at the front. Include actual hard unknot fixtures, determinant-one nontrivial knots, mixed-sign braids of at least four strands, and examples with large intermediate but small final homology. Retain source provenance and the distinction between PD-only and braid-source inputs. The correct metrics include crossings, runs, degree span, maximum layer dimension, cancellation fill, map recompilation, and the compressed context size from research question 10.1. Artificially labeling an easy torus or three-strand example as hard would make this research programme less informative.

10.9 Cost the hierarchy route operation by operation

Research question 10.9. Can the remaining hierarchy steps be given explicit input/output bit contracts and composed into a complete quasi-polynomial runtime proof?

The required ledger includes hierarchy length, boundary-pattern size, surface-coordinate bit length, component tracking after cutting, product-region recognition, and the search cost of each next surface. Polynomial certificate verification must be kept separate from finding the certificate. Compact normal coordinates must remain compact through every update. Lackenby’s current results provide substantial ingredients [13]; completing this ledger would be an independent route, not a corollary of the present Khovanov recurrence.

10.10 Formalize the finite algebraic core first

Research question 10.10. Which parts admit a small proof-assistant development with a stable boundary to the topological literature?

The square-zero calculation, signed interval bookkeeping, rank identity, positive and negative profile formulas, and cycle–cocycle witness criterion are finite algebra and arithmetic. They are natural first targets. A formalization should expose the local tangle-normal-form assumption and then separately formalize or import its topological proof. A machine-checked recurrence conditional on that assumption is valuable, but is not a formalization of the complete unknot detector. The current package contains conventional proofs and executable cross-checks, not a Lean proof.

10.11 A proposed sequence of research milestones

1. Integrate the additive code and preserve the default dispatcher. Reproduce the 254-method suite and the independent cube checks.
2. Build dot- and saddle-compatible context contractions and measure effective amplitude on filter-undecided examples.
3. Prove a complete two-variable theorem, including its chambers, component parities, and compact output.

4. Construct and independently verify compact nontriviality witnesses, with an explicit positive endpoint for unknot cases.
5. Test a representation closed under repeated tangle gluing against the size and operation contract in proposition 9.2.
6. Pursue the hierarchy cost ledger in parallel, so the programme does not depend on one algebraic representation succeeding universally.

The immediate work is concrete and testable. A general quasi-polynomial claim should be made only when both a compact representation and the cost of every operation on it have been proved, including construction and the final decision.

11 Integration, reproducibility, and provenance

11.1 Package layout and proposed repository placement

The archive contains the complete modified `fast/` tree and an untouched `reference/fast/` snapshot from the pinned revision. The integration patch is relative to `Topology/UnknotRecognition/fast`. It adds the modules and tests described below and the optional source-profile argument; it does not alter the default recognition dispatch. The article and research artifacts can be placed in a new report directory under `Topology/UnknotRecognition/reports/`, using the next free report number at integration time.

File or directory	Purpose
<code>article/main.tex</code> and section files	Complete article source, with an ordinary LaTeX bibliography.
<code>article/main.pdf</code>	Compiled and visually checked article.
<code>article/figures/</code>	Vector figures and PNG previews, generated from exact output or retained measurements.
<code>fast/fastunknot/twist/tail.py</code>	Exact one-reference homological recurrence and compact-profile API.
<code>fast/fastunknot/twist/streaming.py</code>	Bounded-cache, degree-streamed full homology and early recognition.
<code>fast/fastunknot/braid_profile.py</code>	Run-encoded structural certificates and the dominated signature experiment.
<code>fast/fastunknot/twist/continuation.py</code>	Raw RLE command-line and Python frontend.
<code>fast/tests/</code>	All inherited and new regression tests.
<code>tools/</code>	Portable audits, benchmarks, figure regeneration, validation, and article build scripts.
<code>results/</code>	Raw measurements, pilot runs, exact comparison records, and the complete test log.
<code>reference/</code>	Pinned baseline and detailed proof/accounting audit notes.
<code>integration.patch</code>	Additive patch for review against the pinned baseline.

The baseline MIT No Attribution (MIT-0) license is retained. No remote repository write or default-backend replacement is part of this deliverable. The accompanying integration guide gives the exact paths and patch check command.

11.2 Interfaces and examples

From the package's `fast/` directory, the new frontend accepts run-encoded JSON:

```
{"strands": 3, "runs": [[1, 1001], [2, -1], [1, 1], [2, -1]]}
```

The corresponding commands are:

```
python -m fastunknot.twist.continuation input.json
python -m fastunknot.twist.continuation input.json \
    --mode homology --method tail --check-d2
python -m fastunknot.twist.continuation input.json \
    --mode homology --method streaming
python -m fastunknot.twist.continuation input.json --mode profile
```

The default is recognition with the tail method selected for any necessary exact endpoint. Recognition first checks the original one-component closure and tries the direct structural certificate. Only an inconclusive certificate reaches the selected backend. Full homology mode always computes the selected full exact homology, including for links; it does not stop after a cheap knot verdict. Profile mode returns only the structural certificate. The methods `tail`, `streaming`, and `macro` are explicit choices.

A Python call avoids JSON serialization limits for extremely large integers:

```
from fastunknot.twist.core import Run
from fastunknot.twist.tail import tail_homology, profile_dimension

m = (1 << 100000) + 1
answer = tail_homology(2, [Run(1, m)], check_d2=True)
assert answer["reduced_rank"] == m
assert answer["reference"]["stats"]["basis"] == 4
assert profile_dimension(answer["degree_profile"], m) == 1
```

The standard Python JSON integer conversion has a decimal-digit limit in this runtime. The API supports the tested 100,001-bit integers directly; the CLI does not disable that general runtime limit. JSON also converts dictionary degree keys to strings. The documentation identifies the schema and inclusive interval semantics so downstream consumers do not confuse a large encoded endpoint with an explicit array index range.

Existing checked diagrams can opt into the source specialization via `recognize(diagram, use_braid_profile=True)` or the production `-braid-profile` option. It is used only with retained checked braid source. A PD-only diagram does not acquire invented braid provenance, and a later diagram factor is not assumed to share the original source braid.

11.3 Reproduction commands

From the archive root:

```
PYTHONPATH=fast python -m unittest discover -s fast/tests -v
python tools/audit_long_twist.py
python tools/audit_slope.py
python tools/benchmark_tail.py
python tools/benchmark_streaming.py --rounds 7 --batch 5
python tools/benchmark_braid_profile.py --rounds 7
python tools/make_figures.py
bash tools/build_article.sh
```

The exact outputs originally used in the article are retained. Rerunning measurements overwrites their default target files; retain a copy or specify a separate output path for comparison. Wall-clock measurements on another machine are expected to differ. The scripts use package-relative imports, fixed stated random seeds, and standard-library exact arithmetic for the core algorithms. Matplotlib is used only to regenerate plots. Building the article requires a conventional LaTeX installation with the packages named in the preamble and `latexmk`.

The independent recurrence audit was executed successfully before being relocated into the final package. The relocated script changes paths and explanatory text, not the tested algebra, and its raw result file is retained unchanged. Its provenance note states this explicitly. The final integrated 254-method suite was run on the delivered combined implementation.

11.4 Evidence boundaries and integration decisions

The proofs establish a theorem for finite complexes obtained from the stated local normal form. The implementation comparisons check that the delivered code computes that model on the tested inputs. Neither is a proof-assistant formalization of the entire topology dependency chain. Benchmarks establish observations on specified input families and execution conditions; they do not establish a distribution-free performance claim.

For integration, the evidence supports adding the tested APIs, research documentation, and opt-in source-profile stage. The streamed backend should remain selectable on memory grounds while its measured time loss is addressed. A future automatic dispatcher should be justified on the subset of inputs reaching the expensive endpoint, with both time and memory objectives stated.

The main unresolved theorem is universal closure of a compact representation under construction, gluing, reduction, and exact decision. The finite-threshold result supplies one such closure mechanism in a controlled setting, together with enough source, proof, and experimental detail to make the next stage of research reviewable and reproducible.

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